



Hello! How is it going?



Agenda

- Recall last session
- Discuss current topic
- Practice
- Quiz

Total Recall



What do you remember most from few previous workshops? (short answer)



How do you install Maven on Windows?

0 ✓

Set PATH variable after
installation

0 ✗

Install via npm

0 ✗

Download source code manually

0 ✗

Create a virtual machine



Where is Maven *project* configuration is stored?

0 ✗

src/main/java

0 ✓

pom.xml file

0 ✗

settings.xml file

0 ✗

application.properties file

Where is Maven *tool* configuration is stored?

0 ✗

src/main/java

0 ✗

pom.xml file

0 ✓

settings.xml file

0 ✗

application.properties file

Maven project structure:

src/

---- main/

---- ---- java/

---- ---- resources/

---- test/

---- ---- java/

---- ---- resources/

target/

pom.xml



Next topic



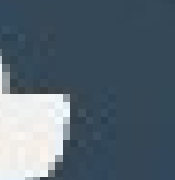


Random fact

Your brain alone burns around 400 to 500 calories each day

Let's explore Maven website shortly:

<https://maven.apache.org/>



Have you looked through the materials?

0



yes

0



no

The Build Lifecycle

Maven terminology:

- **Lifecycle:** The overall sequence of build steps (e.g., default, clean, site).
- **Phase:** A step or milestone in the lifecycle (e.g., compile, test, package). This is the official Maven term for each step in the build process.
- **Goal:** A specific task provided by a plugin that is bound to a phase (e.g., compiler:compile, surefire:test).

1. Lifecycle (Default)

└─ Phases

- | └─ validate
- | └─ Goals/Tasks (e.g., enforcer:enforce)
- | └─ initialize
- | └─ Goals/Tasks (e.g., buildnumber:create)
- | └─ generate-sources
- | └─ Goals/Tasks (e.g., jaxb2:generate)
- | └─ process-sources
- | └─ Goals/Tasks (e.g., antlr:process)
- | └─ generate-resources
- | └─ Goals/Tasks (e.g., templating:generate)
- | └─ process-resources
- | └─ Goals/Tasks (e.g., resources:resources)
- | └─ compile
- | └─ Goals/Tasks (e.g., compiler:compile)

- | process-classes
 - | | Goals/Tasks (e.g., bytecode:process)
- | generate-test-sources
 - | | Goals/Tasks (e.g., testgen:generate)
- | process-test-sources
 - | | Goals/Tasks (e.g., test-antlr:process)
- | generate-test-resources
 - | | Goals/Tasks (e.g., test-templating:generate)
- | process-test-resources
 - | | Goals/Tasks (e.g., resources:testResources)
- | test-compile
 - | | Goals/Tasks (e.g., compiler:testCompile)
- | process-test-classes
 - | | Goals/Tasks (e.g., bytecode:testProcess)
- | test
 - | | Goals/Tasks (e.g., surefire:test)

----- | prepare-package
----- | | Goals/Tasks (e.g., shade:shade)
----- | package
----- | | Goals/Tasks (e.g., jar:jar, war:war)
----- | pre-integration-test
----- | | Goals/Tasks (e.g., failsafe:pre-integration-test)
----- | integration-test
----- | | Goals/Tasks (e.g., failsafe:integration-test)
----- | post-integration-test
----- | | Goals/Tasks (e.g., failsafe:post-integration-test)
----- | verify
----- | | Goals/Tasks (e.g., verify:verify)
----- | install
----- | | Goals/Tasks (e.g., install:install)
----- | deploy
----- | | Goals/Tasks (e.g., deploy:deploy)

However, in most typical Java projects, only a subset of these phases have plugin goals bound to them by default. That's why you often see output for just a few main tasks, such as for `mvn test`:

- resources:resources (process-resources)
- compiler:compile (compile)
- resources:testResources (process-test-resources)
- compiler:testCompile (test-compile)
- surefire:test (test)

If you want to see exactly which plugins and goals are bound to each phase in your project, you can run:

```
mvn help:describe -Dcmd=test -Ddetail
```

or

```
mvn help:describe -Dcmd=test
```


Have you noticed the amount of phases?

0 ✗

5

0 ✓

23

0 ✗

8

0 ✗

47



The default Maven build lifecycle contains **23 phases**. Here is the complete list:

1. validate 2. initialize
3. generate-sources 4. process-sources
5. generate-resources 6. process-resources
7. compile
8. process-classes
9. generate-test-sources 10. process-test-sources
11. generate-test-resources 12. process-test-resources
13. test-compile
14. process-test-classes
15. test
16. prepare-package 17. package
18. pre-integration-test 19. integration-test 20. post-integration-test
21. verify
22. install
23. deploy

basic mvn commands:

- compile
- test-compile
- clean
- test
- package
- install

Maven Plug-ins

- A plugin is a collection of one or more goals.
- A goal is a "unit of work" in Maven.

Maven consists of several core plugins:

- JAR plugin (archive)
- Compiler plugin (compile)
- Surefire plugin (test)
- Create custom plugins

<https://maven.apache.org/plugins/index.html>

Explore. Run. Compare.



Core plugins:

- mvn test-compile
- mvn compiler:testCompile

Packaging types/tools:

- mvn jar:jar
- mvn package

Reporting plugins:

- `mvn javadoc:javadoc`

Tools:

- `mvn help:describe -Dplugin=archetype`

Managing Dependencies

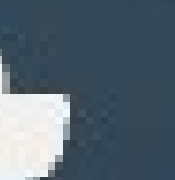
- The junit dependency has the test scope tag - compile is a default.
- It indicates in which lifecycle phase this dependency is used.
- Scopes include: test, compile, provided, runtime, and system.

- <https://central.sonatype.com/search?q=org.junit>
- <https://mvnrepository.com/search?q=org.junit>
- C:\Users\Andrei_Rohau\.m2 (Windows)
- IntelliJ Idea -> External libraries section

Github copilo



QUIZ



What is the default file for Maven project configuration?

0 ✗

build.gradle

0 ✗

settings.xml

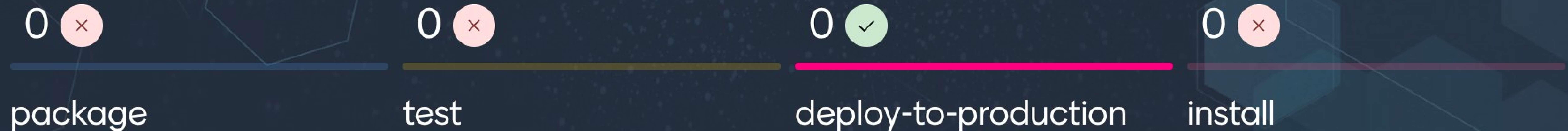
0 ✗

application.yml

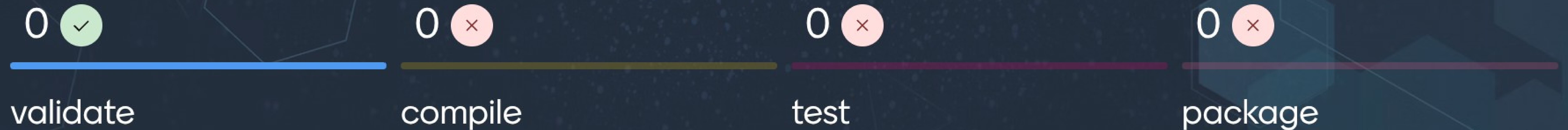
0 ✓

pom.xml

Which is NOT a phase in the default Maven lifecycle?



What is the first phase of a Maven build lifecycle?



What does mvn clean do?

0 ✗

Executes the test phase

0 ✓

Removes previous build
files

0 ✗

Installs third-party plugins

0 ✗

Updates the POM file

Which Maven phase runs jar creation?

0 ✗

test

0 ✗

deploy

0 ✓

package

0 ✗

compile

How are dependencies defined in Maven?

0 ✗

Using dependencies.xml

0 ✗

Manually in source code

0 ✓

Inside the `<dependencies>`
section of POM

0 ✗

During the validate phase

How does Maven resolve dependencies?

0 ✗

Generates dependencies at runtime

0 ✓

Downloads dependencies from repositories

0 ✗

Loads dependencies from a file

0 ✗

By direct linking in .mvn folder

At which phase are integration tests executed?

0 ✓

verify

0 ✗

clean

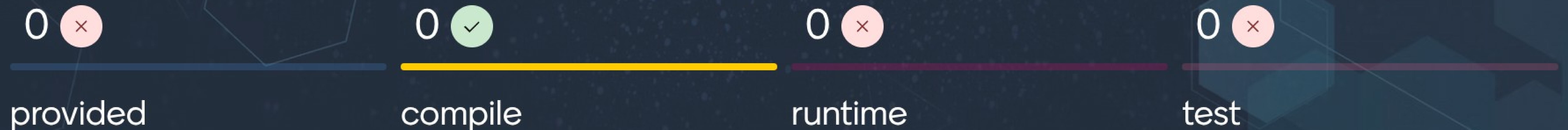
0 ✗

compile

0 ✗

package

What is the default scope for Maven dependencies?



How do you exclude a dependency in Maven?

0 ✗

Remove it from ``$M2_HOME``
folder

0 ✗

Add an override for it

0 ✓

Use ``<exclusions>`` in POM

0 ✗

Manually delete the jar file

Quiz leaderboard

No results yet

Top Quiz participants will be displayed here once there are results!

Q&A part

0 questions
0 upvotes

